AGA KHAN UNIVERSITY EXAMINATION BOARD

HIGHER SECONDARY SCHOOL CERTIFICATE

CLASS XII

ANNUAL EXAMINATIONS (THEORY) 2025

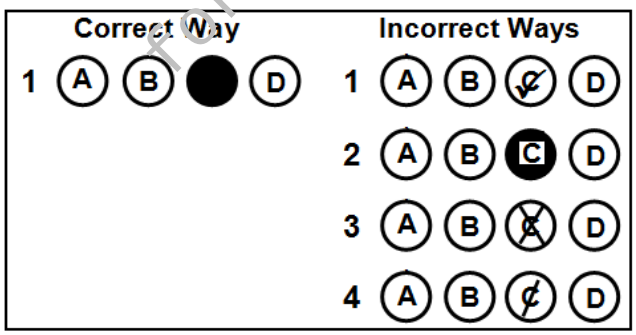
Mathematics Paper I

Time: 1 hour 30 minutes Marks: 50

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INSTRUCTIONS

1. Read each question carefully.
2. Answer the questions on the separate answer sheet provided. DO NOT write your answers on the question paper.
3. There are 100 answer numbers on the answer sheet. Use answer numbers 1 to 50 only.
4. In each question, there are four choices A, B, C, D. Choose ONE. On the answer grid, black out the circle for your choice with a pencil as shown below.



Candidate's Signature

5. If you want to change your answer, ERASE the first answer completely with a rubber, before blacking out a new circle.
6. DO NOT write anything in the answer grid. The computer only records what is in the circles.
7. A formulae list is provided on page 2 and 3. You may refer to it during the paper, if you wish.
8. You may use a scientific calculator if you wish.

List of Formulae

Note:

- The symbols used in the formulae have their usual meaning.

Functions and Limits

$$\lim_{\theta\to 0}\frac{\sin\theta}{\theta}=1 \qquad \lim_{x\to 0}\frac{\sqrt{x+a}-\sqrt{a}}{x}=\frac{1}{2\sqrt{a}} \qquad \lim_{x\to a}\frac{x^n-a^n}{x-a}=na^{n-1}$$

$$\lim_{x\to 0}(1+x)^{\frac{1}{x}}=e \qquad \lim_{x\to+\infty}\left(1+\frac{1}{x}\right)^x=e \qquad \lim_{x\to 0}\frac{a^x-1}{x}=\log_e a$$

Differentiation

$$\frac{\Delta y}{\Delta x}=\frac{f(x+h)-f(x)}{h} \qquad \frac{d}{dx}(\sec x)=\sec x\tan x \qquad \frac{d}{dx}x^n=nx^{n-1} \qquad \frac{d}{dx}(\sin x)=\cos x$$

$$\frac{d}{dx}(\cos x)=-\sin x \qquad \frac{d}{dx}(\tan x)=\sec^2 x \qquad \frac{d}{dx}(\cot x)=-\operatorname{cosec}^2 x \qquad \frac{d}{dx}(\operatorname{cosec} x)=-\operatorname{cosec} x\cot x$$

$$\frac{d}{dx}\left[\sin^{-1}x\right]=\frac{1}{\sqrt{1-x^2}},\ x\in(-1,1) \qquad \frac{d}{dx}\left[\cos^{-1}x\right]=-\frac{1}{\sqrt{1-x^2}},\ x\in(-1,1) \qquad \frac{d}{dx}\left[\tan^{-1}x\right]=\frac{1}{x^2+1},\ x\in R \qquad \frac{d}{dx}\left[\sec^{-1}x\right]=\frac{1}{|x|\sqrt{x^2-1}},\ x\in[-1,1]$$

$$\frac{d}{dx}\left[\operatorname{cosec}^{-1}x\right]=-\frac{1}{|x|\sqrt{x^2-1}},\ x\in[-1,1] \qquad \frac{d}{dx}\left[\cot^{-1}x\right]=-\frac{1}{1+x^2},\ x\in R \qquad \frac{d}{dx}(a^x)=a^x\ln a \qquad \frac{d}{dx}(\log_a x)=\frac{1}{x}\times\frac{1}{\ln a}$$

$$\frac{d}{dx}(\sinh x)=\cosh x \qquad \frac{d}{dx}(\cosh x)=\sinh x \qquad \frac{d}{dx}(\tanh x)=\operatorname{sech}^2 x$$

$$\frac{d}{dx}(\operatorname{cosech} x)=-\coth x\operatorname{cosech} x \qquad \frac{d}{dx}(\operatorname{sech} x)=-\tanh x\operatorname{sech} x \qquad \frac{d}{dx}(\coth x)=-\operatorname{cosech}^2 x$$

$$\frac{d}{dx}\left[f(x)\times g(x)\right]=f(x)\frac{d}{dx}\left[g(x)\right]+g(x)\frac{d}{dx}\left[f(x)\right] \qquad \frac{d}{dx}\left(\frac{f(x)}{g(x)}\right)=\frac{g(x)\frac{d}{dx}\left[f(x)\right]-f(x)\frac{d}{dx}\left[g(x)\right]}{\left[g(x)\right]^2}$$

Maclaurin Series $f(x)=f(0)+f'(0)x+\frac{f''(0)}{2!}x^2+\frac{f'''(0)}{3!}x^3+\ldots+\frac{f^n(0)}{n!}x^n+\ldots$

Taylor's Series $f(x)=f(a)+f'(a)(x-a)+\frac{f''(a)}{2!}(x-a)^2+\frac{f'''(a)}{3!}(x-a)^3+\frac{f^{iv}(a)}{4!}(x-a)^4+\ldots+\frac{f^n(a)}{n!}(x-a)^n+\ldots$

Integration

$$\int f'(x)dx=f(x)+c \qquad \int[f(x)]^n f'dx=\frac{[f(x)]^{n+1}}{n+1}+c,\quad (n\neq -1) \qquad \int f(x)g'(x)dx=f(x)g(x)-\int f'(x)g(x)dx$$

$$\int\frac{f'(x)}{f(x)}dx=\ln f(x)+c \qquad \int\operatorname{cosec} x\,dx=\ln|\operatorname{cosec} x-\cot x|+c \qquad \int\sec x\,dx=\ln|\sec x+\tan x|+c$$

Plane Analytical Geometry (Straight Line)

$$d=\frac{|ax_1+by_1+c|}{\sqrt{a^2+b^2}} \qquad y-y_1=\frac{y_2-y_1}{x_2-x_1}(x-x_1) \qquad \left(\frac{x_1+x_2}{2},\frac{y_1+y_2}{2}\right)$$

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Point of internal division

$$\left(\frac{k_1x_2+k_2x_1}{k_1+k_2}, \frac{k_1y_2+k_2y_1}{k_1+k_2}\right)$$

Point of external division

$$\left(\frac{k_1x_2-k_2x_1}{k_1-k_2}, \frac{k_1y_2-k_2y_1}{k_1-k_2}\right)$$

$$\frac{x-x_1}{\cos\alpha}=\frac{y-y_1}{\sin\alpha}=r \text{ (say)}$$

$$d=\sqrt{(x_2-x_1)^2+(y_2-y_1)^2}$$

$$\frac{x}{a}+\frac{y}{b}=1$$

$$x\cos\alpha+y\sin\alpha=p$$

$$y=mx+c$$

$$y-y_1=m(x-x_1)$$

$$\theta=\tan^{-1}\left[\frac{m_2-m_1}{1+m_1m_2}\right]$$

$$\theta=\tan^{-1}\left[\frac{2\sqrt{h^2-ab}}{a+b}\right]$$

$$\begin{vmatrix} a_1 & b_1 & c_1 \\ a_2 & b_2 & c_2 \\ a_3 & b_3 & c_3 \end{vmatrix}=0$$

$$\Delta=\frac{1}{2}\begin{vmatrix} x_1 & y_1 & 1 \\ x_2 & y_2 & 1 \\ x_3 & y_3 & 1 \end{vmatrix}$$

Circles

$$(x-h)^2+(y-k)^2=r^2$$

$$x^2+y^2+2gx+2fy+c=0$$

Equation of normal to a circle

$$(y-y_1)(x_1+g)=(x-x_1)(y_1+f)$$

Equation of tangent to a circle

$$xx_1+yy_1+g(x+x_1)+f(y+y_1)+c=0$$

Length of tangent to a circle from a point (x_1, y_1), $l=\sqrt{x_1^2+y_1^2+2gx_1+2fy_1+c}$

Parabola

$x^2=4ay$ $\quad (x-h)^2=4a(y-k)$ $\quad y^2=4ax$ $\quad (y-k)^2=4a(x-h)$

Ellipse

$\frac{(x-h)^2}{a^2}+\frac{(y-k)^2}{b^2}=1$, $a>b$ $\quad b^2=a^2(1-e^2)$ $\quad c=ae$ $\quad \frac{(y-k)^2}{a^2}+\frac{(x-h)^2}{b^2}=1$, $a>b$

Hyperbola

$\frac{(x-h)^2}{a^2}-\frac{(y-k)^2}{b^2}=1$ $\quad b^2=a^2(e^2-1)$ $\quad c=ae$ $\quad \frac{(y-k)^2}{a^2}-\frac{(x-h)^2}{b^2}=1$

Translation and Rotation

$x=X+h$, $y=Y+k$ $\quad X=x\cos\theta+y\sin\theta$, $Y=y\cos\theta-x\sin\theta$ $\quad \tan 2\theta=\frac{2h}{a-b}$

Vectors

$\underline{u}.\underline{v}=|\underline{u}||\underline{v}|\cos\theta$ $\quad$ Area of a triangle $=\frac{1}{2}|\underline{u}\times\underline{v}|$ $\quad \underline{u}\times\underline{v}=\left(|\underline{u}||\underline{v}|\sin\theta\right)\hat{\underline{n}}$ $\quad \underline{r}=\frac{q\underline{a}+p\underline{b}}{p+q}$

$$\left|\overrightarrow{P_1P_2}\right|=\sqrt{(x_2-x_1)^2+(y_2-y_1)^2+(z_2-z_1)^2}$$

$$\frac{1}{6}(\underline{u}\times\underline{v}).\underline{w}=\frac{1}{6}[\underline{u}\ \underline{v}\ \underline{w}]$$

$$\underline{u}.(\underline{v}\times\underline{w})=\begin{vmatrix} a_1 & b_1 & c_1 \\ a_2 & b_2 & c_2 \\ a_3 & b_3 & c_3 \end{vmatrix}$$

Some Trigonometric Identities

$$\cos(\alpha\pm\beta)=\cos\alpha\cos\beta\mp\sin\alpha\sin\beta$$

$$\sin(\alpha\pm\beta)=\sin\alpha\cos\beta\pm\cos\alpha\sin\beta$$

$$\cos P+\cos Q=2\cos\frac{P+Q}{2}\cos\frac{P-Q}{2}$$

$$\cos P-\cos Q=-2\sin\frac{P+Q}{2}\sin\frac{P-Q}{2}$$

$$\sin P+\sin Q=2\sin\frac{P+Q}{2}\cos\frac{P-Q}{2}$$

$$\sin P-\sin Q=2\cos\frac{P+Q}{2}\sin\frac{P-Q}{2}$$

$$\sin 2\alpha=2\sin\alpha\cos\alpha$$

$$\cos 2\alpha=\cos^2\alpha-\sin^2\alpha$$

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1. The domain of $g(x)=\frac{2x-3}{x^2-a^2}$ is

   A. $R-\{a, -a\}$
   B. $R-\{a^2, -a^2\}$
   C. $R-\{a, -a, 3\}$
   D. $R-\{a^2, -a^2, 3\}$

2. Which of the following equations exemplifies a linear function?

   A. $y=\frac{3}{2x+1}$
   B. $y=\frac{2x+1}{3}$
   C. $y=2x^{-2}+1$
   D. $y=2x(3x+1)$

3. The value of $\lim_{x\to 0}\left[\frac{\sin 2x}{3x}\right]$ is

   A. $\frac{1}{3}$
   B. $\frac{2}{3}$
   C. 2
   D. ∞

4. If $f(x)=\begin{cases}1-2x & x>1\\ 3x-2 & x\le 1\end{cases}$, then $\lim_{x\to 1} f(x)$

   A. is -1
   B. is 0
   C. is 1
   D. does not exist

5. If $y=\frac{3}{2x+1}$, then $\frac{dy}{dx}$ will be

   A. $\frac{3}{2}$
   B. $\frac{3}{2x^2}$
   C. $-3(2x+1)^{-2}$
   D. $-6(2x+1)^{-2}$

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6. The derivative of $x(x-2)^5$ is

A. $5(x-2)^4$
B. $5x(x-2)^4$
C. $5(x-2)^4 + (x-2)^5$
D. $5x(x-2)^4 + (x-2)^5$

7. If $y = \frac{x}{1+\pi x}$, then $\frac{dy}{dx}$ will be

A. 1
B. $\frac{1}{\pi}$
C. $\frac{1}{(1+\pi x)^2}$
D. $\frac{1+2\pi x}{(1+\pi x)^2}$

8. On differentiating $f(t) = e^{-et}$ with respect to t, we get

A. e^{-e}.
B. e^{-et}.
C. $-e^{-et}$.
D. $-e \times e^{-et}$.

9. If $g(t) = \log_3(1+x)$, then $g'(t)$ will be

A. $\frac{1}{x+1}$
B. $\frac{1}{x+1} \times \ln(x+1)$
C. $\frac{1}{x+1} \times \ln 3$
D. $\frac{1}{x+1} \times \frac{1}{\ln 3}$

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10. If $y = p\tan^{-1} x$, then $\frac{dy}{dx}$ is equal to

A. $\frac{p}{1+x^2}$

B. $\frac{p}{\sec^2 x}$

C. $\frac{1}{p\sec^2 x}$

D. $\frac{1}{p(1+x^2)}$

11. A ball is projected upwards, that moves in a trajectory as shown in the given diagram.



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If the equation for displacement at any point 't' is expressed as $y = \frac{49}{10}t^2$, then the acceleration of the ball after 15 seconds will be

A. 1102.5 m/s^2
B. 147 m/s^2
C. 73.5 m/s^2
D. 9.8 m/s^2

12. The equations of tangent and normal to a curve, at a certain point, are presented in the given table.

Equation of Tangent	Equation of Normal
$y = 4x - 3$	$y = ax + \frac{11}{2}$

The value of *a* is

A. -4

B. $-\frac{1}{4}$

C. $\frac{1}{4}$

D. 4

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13. The fraction $\frac{1}{x\left(x^2-3x+2\right)}$ can be resolved into partial fraction as

A. $\frac{A}{x}+\frac{B}{x^2-3x+2}$

B. $\frac{A}{x}+\frac{Bx+C}{x^2-3x+2}$

C. $\frac{A}{x}+\frac{B}{(x-2)(x-1)}$

D. $\frac{A}{x}+\frac{B}{x-2}+\frac{C}{x-1}$

14. If the derivative of a function is $f'(x)=e^{ax}$, then the function $f(x)$ will be

A. $e^{-ax}+C$.

B. $e^{ax}+C$.

C. $\frac{e^{ax}}{a^{-1}}+C$.

D. $\frac{e^{ax}}{a}+C$.

15. The integral $\int \frac{\sin^2 x}{\sec x}dx$ is equal to

A. $-\frac{\cos^3 x}{3}+C$.

B. $\frac{\sin^3 x}{3}+C$.

C. $-\cos^3 x+C$.

D. $\sin^3 x+C$.

16. The integral $\int xe^{ax}dx$ is equal to

A. $x\times\frac{e^{ax}}{a}-\int e^{ax}dx+C$.

B. $x\times\frac{e^{ax}}{a}-\frac{1}{a}\int e^{ax}dx+C$.

C. $\frac{x^2e^{ax}}{2}-\int\frac{x^2}{2}\times e^{ax}dx+C$.

D. $\frac{x^2e^{ax}}{2}-\int\frac{x^2}{2}\times\frac{e^{ax}}{a}dx+C$.

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17. The value of $\int_0^{\pi} \sin \frac{x}{2} dx$ will be

A. -4
B. -2
C. 2
D. 4

18. If $\int_0^{\pi} k(x)dx = -\frac{3\pi}{2}$, then the value of $\int_{\pi}^{0} [k(x)+1]dx$ is

A. $-\frac{3\pi}{2}+1$
B. $\frac{3\pi}{2}+1$
C. $\frac{\pi}{2}$
D. -2π

19. The solution of the differential equation $\frac{dy}{dx} = \frac{1}{y(x+1)}$ is

A. $\ln\left(\frac{y}{x+1}\right) = C$.
B. $y = \ln(x+1) + C$.
C. $\frac{y^2}{2} = \ln(x+1) + C$.
D. $y = \ln y + \ln(x+1) + C$.

20. A ball rolls in a straight line such that *t* seconds after passing through a fixed point *O*, its displacement is given by $3t + t^2$ m.

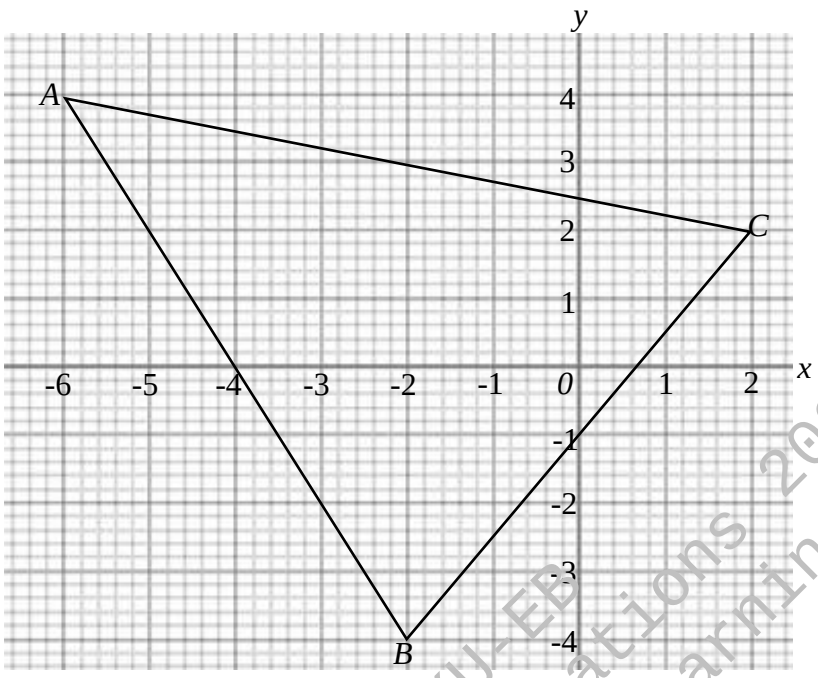
The velocity of the ball is given by

A. $t + 2t$ ms^{-1}
B. $3 + 2t$ ms^{-1}
C. $6t^2 + 3t^3$ ms^{-1}
D. $\frac{3}{2}t^2 + \frac{t^3}{3}$ ms^{-1}

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Use the given figure to answer Q.21 and Q.22.

(**Note**: Vertices of the triangle *ABC* are shown in the Cartesian coordinate system.)



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21. The length of the median from the vertex *C* to *AB* will be

A. $\sqrt{80}$ unit.
B. $\sqrt{40}$ unit.
C. $\sqrt{32}$ unit.
D. $\sqrt{8}$ unit.

22. If the equation of the side *BC* is $3x - 2y - 2 = 0$, then the shortest distance from the vertex *A* to the line *BC* will be

A. $\frac{28}{\sqrt{13}}$ unit.
B. $\frac{8}{\sqrt{13}}$ unit.
C. $\frac{28}{\sqrt{5}}$ unit.
D. $\frac{8}{\sqrt{5}}$ unit.

23. The slope of a line P is $\frac{a}{b}$, where $(a, b \in N)$.

If a line Q is perpendicular to P, then the slope of Q will be equal to

A. $-\frac{a}{b}$

B. $-\frac{b}{a}$

C. $\frac{a}{b}$

D. $\frac{b}{a}$

24. The slope intercept form of the equation $y - a = \frac{1}{a}(x - a)$ is

A. $y = ax - 1 + a.$

B. $y = ax - 1 - a.$

C. $y = \frac{1}{a}x - 1 - a.$

D. $y = \frac{1}{a}x - 1 + a.$

25. Which of the following points lies on the line $y = -2x - 2$?

A. $(0, 0)$

B. $(0, -2)$

C. $(1, 0)$

D. $(0, 1)$

26. If $l_1 : 2x + y - 3 = 0$ and $l_2 : x - y + 3 = 0$ are two intersecting lines, then the family of lines passing through their intersecting point will be represented by

I. $(2x + y - 3) + 2(x - y + 3) = 0$

II. $(2x + y - 3) - 5(x - y + 3) = 0$

III. $(2x + y - 3) \times (x - y + 3) = 0$

A. I only.

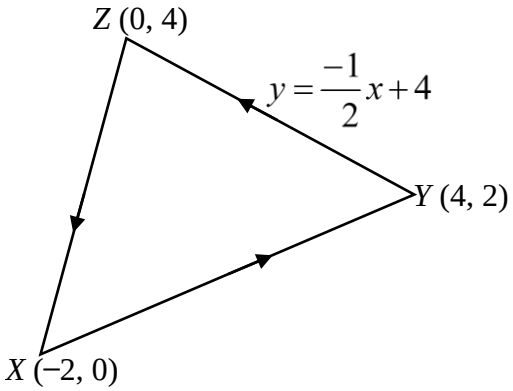
B. III only.

C. I and II.

D. II and III.

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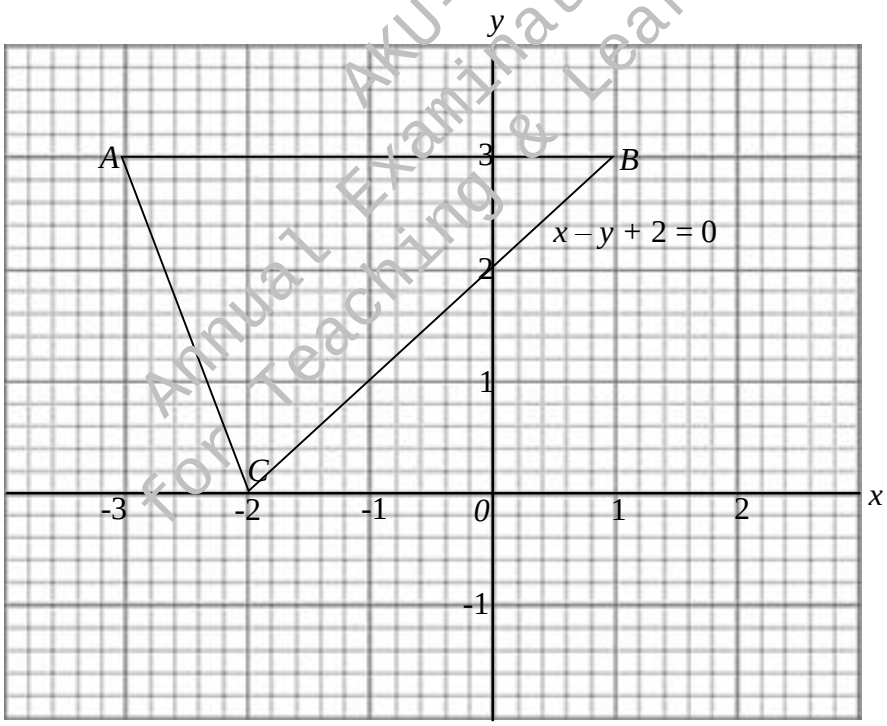
27. In the given triangle, the measure of angle *Y* will be



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A. 135°
B. 120°
C. 60°
D. 45°

28. In the given triangle *ABC*, the equation of the side *BC* is $x - y + 2 = 0$.



The equation of the altitude from vertex *A* to the side *BC* will be

A. $x + y = 0$
B. $x - y = 0$
C. $x + y - 6 = 0$
D. $x - y - 6 = 0$

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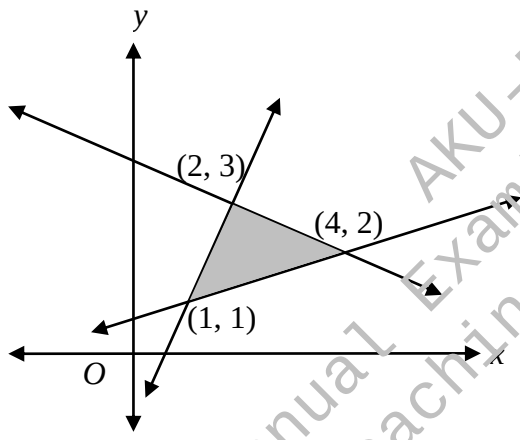
29. Which of the following linear equations is homogenous?

A. $2x-3y+4=0$
B. $2x-y=0$
C. $2x-5=0$
D. $y-5=0$

30. The measure of an acute angle between the lines represented by their joint equation $\frac{1}{4}x^2-y^2=0$, will be

A. $36.86°$
B. $53.13°$
C. $60.79°$
D. $71.45°$

31. The maximum value of $f(x,y)=2x+y$, subject to the constraints shown in the given figure is



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A. 3
B. 7
C. 10
D. 12

32. The radius of a circle represented by the equation $x^2+y^2-4x+3y-5=0$ will be

A. 1.12 unit.
B. 2.82 unit.
C. 3.35 unit.
D. 4.24 unit.

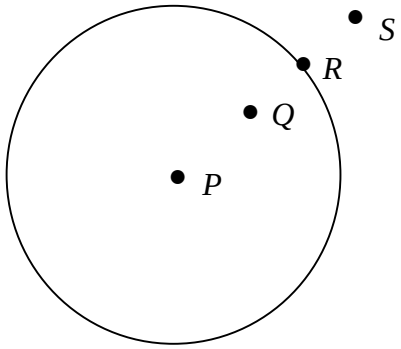
33. The equation of the normal to the circle $x^2+y^2-25=0$ at $(-3,-4)$ will be

A. $4x-3y=0$
B. $3x+4y=0$
C. $3x-4y=0$
D. $4x+3y=0$

34. If (0, −3) is a point that lies on a circle, then the value of *h* in the equation $x^2 + y^2 - 4x + 4y + h = 0$ will be

A. -21
B. -3
C. 3
D. 21

35. The given figure shows a circle $x^2 + y^2 = 40$ with centre at origin (*P*).



If $x = 4$ and $y = 5$, then the point that represents the location is

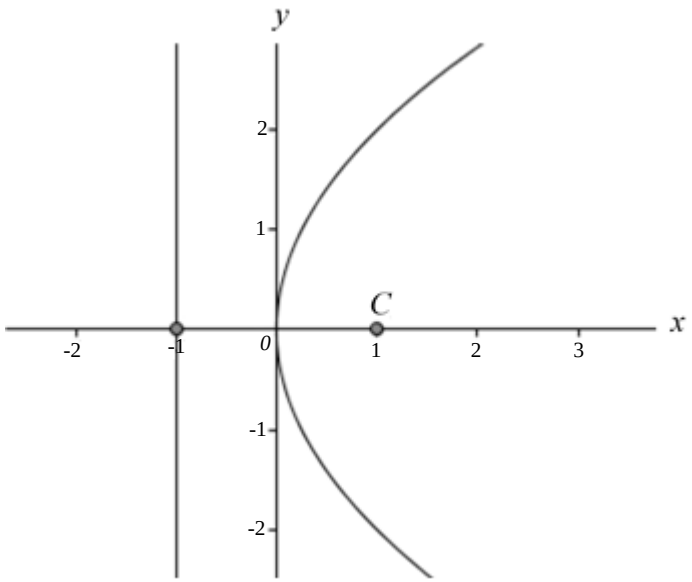
A. *P*.
B. *Q*.
C. *R*.
D. *S*.

36. For the equation of parabola $y^2 = 4ax$, the slope of tangent at (x_1, y_1) is

A. $\frac{4a}{x_1}$
B. $\frac{4a}{y_1}$
C. $\frac{2a}{x_1}$
D. $\frac{2a}{y_1}$

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Use the given graph of a standard parabola to answer Q.37 and Q.38.



37. The equation of the given parabola will be

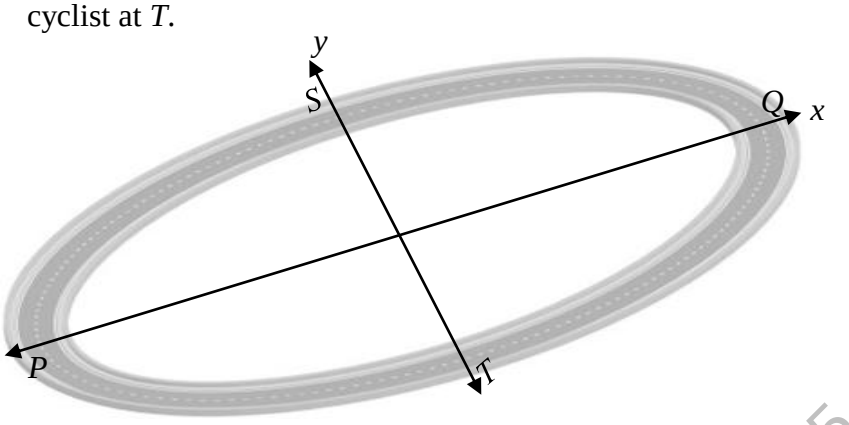
A. $x^2 = -4y$.
B. $y^2 = -4x$.
C. $x^2 = 4y$.
D. $y^2 = 4x$.

38. The length of latus rectum of the given parabola will be

A. 1
B. 2
C. 3
D. 4

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39. There is an elliptic cycling track around a park. While cycling, the maximum (direct) distance between two cyclists is 300 m. A cyclist at point *S* is at a (direct) distance of 120 m from a cyclist at *T*.



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The equation for the elliptic cycling track will be

A. $\frac{x^2}{240^2}+\frac{y^2}{300^2}=1$

B. $\frac{x^2}{120^2}+\frac{y^2}{300^2}=1$

C. $\frac{x^2}{150^2}+\frac{y^2}{60^2}=1$

D. $\frac{x^2}{300^2}+\frac{y^2}{120^2}=1$

40. The equation of an ellipse is given as $\frac{(y-2)^2}{25}+\frac{(x+2)^2}{16}=1$.

If the foci lies at $(-2,\ 2\pm 3)$, then the equation of directrix will be

A. $x=-2\pm\frac{25}{3}$

B. $y=-2\pm\frac{25}{3}$

C. $x=2\pm\frac{25}{3}$

D. $y=2\pm\frac{25}{3}$

41. The equation of tangent to the ellipse $\frac{x^2}{15}+\frac{y^2}{10}=1$ at (3, 2) will be

A. $x+y-5=0$

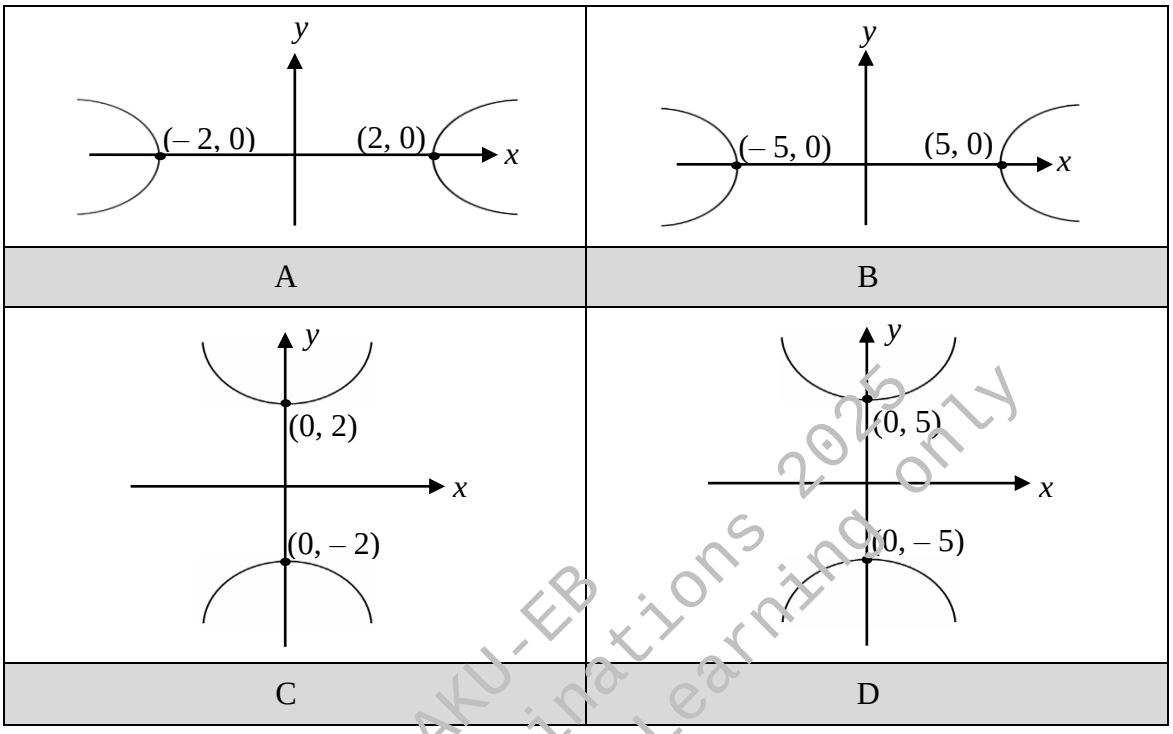
B. $x-y-1=0$

C. $x+y-6=0$

D. $x+y+1=0$

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42. Which of the following graphs represents the hyperbola $\frac{y^2}{4} - \frac{x^2}{25} = 1$?



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43. The equation for the transverse axis of the hyperbola $(x-1)^2 - 4y^2 = 4$ will be

A. $x = 1$
B. $y = 0$
C. $x = 0$
D. $x = -1$

44. The co-ordinates of a point with respect to x y-axes P are (0, 1).

If the axes are rotated anticlockwise about the origin through an angle of 30º, then the y co-ordinate of the new axes of the point P is

A. sin 30°
B. cos 30°
C. sin 60º
D. cos 60º

45. The unit vector in the direction of vector $4\hat{i} - 3\hat{j}$ is

A. $\hat{i} - \hat{j}$.
B. $\hat{i} + \hat{j}$.
C. $\frac{4}{5}\hat{i} - \frac{3}{5}\hat{j}$.
D. $\frac{4}{\sqrt{7}}\hat{i} - \frac{3}{\sqrt{7}}\hat{j}$.

46. If $\vec{MN} = -3\hat{i} + 2\hat{j} + \hat{k}$ and N is a point whose position vector is $3\hat{i} + \hat{j} - 2\hat{k}$, then the position vector of M will be

A. $3\hat{j} - \hat{k}$.
B. $\hat{j} - 3\hat{k}$.
C. $6\hat{i} - \hat{j} - 3\hat{k}$.
D. $6\hat{i} + \hat{j} - 3\hat{k}$.

47. Two points in the space are given by $A(2, 1, 0)$ and B (1, 0, 2).

The distance between A and B will be

A. $\sqrt{6}$ unit.
B. $\sqrt{10}$ unit.
C. 6 unit.
D. 10 unit.

48. Consider two vectors $\vec{a} = \hat{i} + p\hat{k}$ and $\vec{b} = 4\hat{j} - \hat{k}$.

If $\vec{a}.\vec{b} = 1$, then the value of p will be

A. -3
B. -1
C. 1
D. 3

49. Consider that, $\vec{u}$ and $\vec{v}$ are NOT null vectors but their cross product is a null vector.

The angle between the vectors $\vec{u}$ and $\vec{v}$ must be

A. 0°
B. 45°
C. 60°
D. 90°

50. If $\vec{A} = \hat{i} - 2\hat{j}$, $\vec{B} = a\hat{i} + \hat{j} + \hat{k}$ and $\vec{C} = \hat{i} + \hat{j} - 2\hat{k}$ are three co-planar vectors, then the value of a will be

A. $-\frac{5}{4}$
B. $-\frac{1}{4}$
C. $\frac{1}{4}$
D. $\frac{5}{4}$

END OF PAPER

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